Bank Balance Sheet Risk Allocation

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Abstract

We formulate the optimal balance sheet management problem as a linear program and study it using a duality approach. In addition to helping to determine the optimal balance sheet, the dual problem also provides us the market prices of interest rate risk and credit risk.

Our methodology is used to determine premia on credit risk and interest rate risk for commercial banks, which in turn allows us to manage the risk allocation for a bank given a risk budget.

Moreover, our approach will be of interest to regulators, who can use it to assess the market price of credit and interest rate risk at each point in the economic cycle.

Finally, we apply this methodology to real data and show how it can be used to a real setting, using diversification constraints and a greedy algorithm that results in the optimal asset- liability allocation.

Key words. Interest rate risk, credit risk, balance sheet optimization, risk allocation, duality.

MSC classification. 52A41, 90C25, 91G99.

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1 Introduction

Bank balance sheet optimization models have seen many advances recently. The literature, which started in the seventies, has recently integrated many aspects, such as regulation, risk constraints, and diverse type of risks. In these models, however, there is a missing link, in particular to know how the balance sheet asset-liability allocation relates to the price of each of the risks the bank incurs.

Knowing the price of each of the risks is important for several reasons: first, the decision maker will want to know which risk pays better, so that he can tilt his risk budget towards the risks that compensate more in terms of expected returns. For example, if interest rate risk pays better than credit risk, the decision maker may want to take on more interest rate risk than credit risk.

Given a risk budget, that encompasses both credit and interest rate risks, a bank may want to decide how to allocate the risk in an efficient way. Should it allocate more risk to credit risk or to interest rate risk? The answer depends on the price of each risk. This is known as the risk allocation problem. No link has been established yet between optimal balance sheets and the risk allocation.

Also, regulators, at each point in the cycle, may want to identify the prices of interest rate risk and credit risk involved in commercial banking asset classes, so that they can address the likelihood of a bubble for instance. Our research bridges the gap between the balance sheet allocation problem and the pricing of each of the risk components, by using a linear programming model along with duality. Moreover, we show how our approach can be used in practice to construct balance sheets using a greedy algorithm.

Recently, there has been a series of research on optimal bank balance sheet management, which we list below:

- Birge and Júdice (2013) calculate the optimal balance sheet in a simplified setting for a one-period model considering credit risk, but they do not consider interest rate risk.

- Halaj (2013) devises an optimal asset structure for a bank, but does not make the connection with risk allocation. Interest rate risk is measured only from the point of view of net interest income risk within a year, and he does not consider the long-run interest rate risk.

- Halaj (2015) considers a balance sheet optimization, but does not consider interest rate risk.

- Schmaltz et. al (2014) consider a balance sheet optimization problem for a non-compliant bank in Basel III to comply with Basel III and does not consider interest rate risk.

- Coelho, Santos e Júdice (2019) consider a robust bank asset structure problem, but only consider the sensitivity to net interest margin, no long-term interest rate risk.

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- Dewasurendra, Júdice and Zhu (2019) consider the optimal balance sheet problem assuming that the bank can leverage on wholesale liabilities, and calculate the optimal interest-rate risk taking from a book-value accounting perspective. The paper does not reflect credit risk and the allocation assumes only four instruments: mortgages, deposits, equity and wholesale funding.

  One empirical paper is important to note:

- Begenau, Piazzesi and Schneider (2015) conduct empirical work to determine the credit and interest rate risk profile of banks. Their methodology consists in building two factor portfolios (one for interest rate risk, comprised of Treasury bonds, and one for credit risk, consisting in BBB- rated bonds). Through the balance sheet and maturity data of bonds and loans, they map each bank's balance sheet to these factor portfolios. For the case of loans, where this approach is the most difficult, they assume that each loan's credit risk is determined by the regulatory risk weight (for example, a loan with 50% risk weight is assumed to have a credit risk equivalent to an A-corporate bond). This approach has a major simplification: a regulatory risk weight does not determine necessarily the risk on the asset, and assets with significantly different levels of credit risk can have the same regulatory risk weight.

With this literature review in mind, our contributions are the following:

- Using a linear programming model and its duality, we establish a link between the balance sheet optimization problem and the pricing of interest rate and credit risk. We also derive explicit procedures for the optimal balance sheet given limits on interest and credit risks that are easy to implement.

- We show that these prices of credit and interest rate risk correspond to no-arbitrage prices of insurance on each of these risks.

- Our methodology is used to determine interest risk and credit risk premia for commercial banks.

- We also show how to use the market prices of each of the risks so that bank can conduct its risk allocations, allocating more to the risks that give better payoffs.

- We finally show how to use this approach in practice, with real data, using diversification constraints, through a greedy algorithm.

The paper is organized as follows: Section 2 develops the linear programming model; Section 3 finds the solution to the problem, using linear programming duality; Section 4 demonstrates that the dual variables correspond to the market prices that insure credit risk and interest rate risk; in Section 5, we apply the methodology using real data; Section 6 relates the balance sheet allocation problem to the risk allocation problem; Section 7 discusses diversification constraints that prevent concentration in asset classes; the conclusion is in section 8.

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Asset	allocation	returns	interest risk	credit risk
General	x_k	r_k	i_k	c_k
Bonds	y_j	R_j	I_j	none
Fed fund	y_0	R_0	none	none

Table 1: Notation

2 A linear programming model

In this section, we form a linear programming model for the risk allocation problem.

2.1 Model specification

We consider a balance sheet that consists of general investable assets such as mortgages, corporate credits, consumer credits and bonds. For the purpose of balance sheet optimization, the general assets are characterized by an expected return, an interest risk and a credit risk per unit of investment. Similarly, bonds are characterized by a return and an interest risk without credit risk. We use notation given in Table 1. For bonds, we use subscript $j > 0$ for long bonds and $j < 0$ for short bonds. Distinguishing long and short bonds is practically important because for the purpose of the balance sheet, the cost of funding for the bank is generally higher than the credit risk-free rate at which the bank can invest. Therefore, the specification for the long and short bonds are different. Subscript $j = 0$ is reserved to the overnight central bank borrowing which has no interest risk ($I_0 = 0$).

For bonds, the interest rate risk is determined by formula

$$I_j = D_j * N^{-1}(0.99) * \sigma_i, \tag{2.1}$$

where the modified duration D_j can be approximated by

$$D_j(y) = \frac{1}{y(y+1)^t} - \frac{1}{y}, \tag{2.2}$$

and σ_i is the annual volatility of interest rates, y is the yield and t is the maturity of the bond.

The interest rate risk is essentially the loss in market value for the balance sheet of the bank in case there is adverse interest rate shock of $N^{-1}(0.99) * \sigma_i$, so one can see it as a Value-at-Risk measure for interest-rate risk.

For general assets, we need to model their credit risk. One of the standards is the Vasicek (2002) model for credit risk, which has been the basis for capital-setting standards in the advanced-IRB approach in Basel II. This model has become popular in its book-value approach for modeling credit losses, i.e., credit losses are usually modeled in a one-year horizon using typically an incurred loss approach, typical in book-value accounting.

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For our purpose, we need to model losses in a mark-to-market approach, so that we have a risk measurement consistent with interest rate risk, which is modeled with a mark-to-market value approach.

Nevertheless, the Vasicek (2002) paper derives the mark-to-market loss distribution for a portfolio of credits. Assume that H is the investment horizon and T is the maturity of the credits. This loss distribution is given by

$$P\left[L \leq x\right] = F\left(\frac{x}{a}; PD; \frac{\rho H}{T}\right), \tag{2.3}$$

$$\tag{2.4}$$

where

$$F(x, p, \rho) = N\left(\frac{\sqrt{1-\rho}N^{-1}(x) - N^{-1}(PD)}{\rho},\right) \tag{2.5}$$

and

$$a = LGD * exp(-r(T-H)). \tag{2.6}$$

Here, LGD is the loss given default and PD is the T-year probability of default, which can be inferred from the marginal probability of default mp

$$PD = 1 - exp(-mp\,T).$$

Based on this loss distribution, we can derive a CreditVaR measure of credit risk, which is given by

$$LGDx^*, \tag{2.7}$$

where x^* solves

$$P[L \leq x^*] = \alpha. \tag{2.8}$$

Let us assume that $H = 1$. Solving 2.8 and plugging in 2.7, we get the number for the CreditVaR based on the Vasicek model

$$a\,N\left(\frac{\sqrt{\frac{\rho}{T}}N^{-1}(\alpha) + N^{-1}(PD)}{\sqrt{1-\frac{\rho}{T}}}\right). \tag{2.9}$$

2.2 Linear programming model

We use notation $sgn(j) = 1$ for $j > 0$ and -1 otherwise. The upper limits of interest rate risk and credit risk are denoted i and c, respectively. We use e to denote the total bank equity. Let us assume that there are l long bonds, s short bonds and n general assets. Then we have the following linear programming problem

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$$
\begin{aligned}
p = \max \quad & \textstyle\sum_{k=1}^{n} r_k x_k + \sum_{j=-s}^{l} sgn(j) R_j y_j \\
\text{subject to} \quad & \\
& \textstyle\sum_{k=1}^{n} i_k x_1 + \sum_{j=-s}^{l} sgn(j) I_j y_j \le i \\
& \textstyle\sum_{k=1}^{n} c_k x_k \le c \\
& \textstyle\sum_{k=1}^{n} x_k + \sum_{j=-s}^{l} sgn(j) y_j = e
\end{aligned}
\tag{2.10}
$$

In this formulation, we assume no short selling is allowed. The bonds that are assets are long bonds signaled by indices $j > 0$, and the bonds in the liabilities are short bonds signaled by indices $j < 0$. The difference is taken care of by using the sign of the index j. The only exception is the central bank funding, y_0 can take both positive and negative values. All the other quantities x_k and y_j are nonnegative.

Substitute

$$
y_0 = \sum_{k=1}^{n} x_k + \sum_{j=1}^{l} y_j - \sum_{j=-s}^{-1} y_j - e
$$

we can rewrite the above problem as

$$
\begin{aligned}
p = \max \quad & \textstyle\sum_{k=1}^{n} (r_k - R_0) x_k + \sum_{j=-s}^{l} sgn(j)(R_j - R_0) y_j + R_0 e \\
\text{subject to} \quad & \\
\lambda_i \qquad & \textstyle\sum_{k=1}^{n} i_k x_k + \sum_{j=-s}^{l} sgn(j) I_j y_j \le i \\
\lambda_c \qquad & \textstyle\sum_{k=1}^{n} c_k x_k \le c \\
& x_k, y_j \ge 0.
\end{aligned}
\tag{2.11}
$$

We observe that the only role equity e plays in this model is to increase the expected return by $R_0 e$. Financially, this means that to the extent that equity is available the bank will fund the operation first using equity and then borrowing from the central bank. In doing so the bank can increase the expected return by $R_0 e$. In practice the size of the bank equity comparing to the funding needs is relatively small, so that it is always used up this way. Thus, in the analysis below we will consider the case $e = 0$ to simplify the notation and the linear programming problem becomes

$$
\begin{aligned}
p = \max \quad & \textstyle\sum_{k=1}^{n} (r_k - R_0) x_k + \sum_{j=-s}^{l} sgn(j)(R_j - R_0) y_j \\
\text{subject to} \quad & \\
\lambda_i \qquad & \textstyle\sum_{k=1}^{n} i_k x_k + \sum_{j=-s}^{l} sgn(j) I_j y_j \le i \\
\lambda_c \qquad & \textstyle\sum_{k=1}^{n} c_k x_k \le c \\
& x_k, y_j \ge 0.
\end{aligned}
\tag{2.12}
$$

The two constraints in (2.12) correspond to dual variables λ_i, λ_c representing the sensitivity with respect to changes in the limit of interest risk and credit risks, respectively.

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We also note that writing the sum $\sum_{j=-s}^{l}$ is for the conciseness of the notation. In fact, the term corresponding to $j = 0$ is 0 in both the return function and the interest rate risk constraint. Therefore, y_0 term is absent.

The linear programming problem (2.12) has a dual problem (the corresponding primal variables are marked in front of the constraints):

$$\begin{array}{rcl} d = \min & & i\lambda_i + c\lambda_c \\ \text{subject to} & & \\ x_k & i_k\lambda_i + c_k\lambda_c \geq r_k - R_0, k = 1, \ldots, n & \\ y_j & sgn(j)I_j\lambda_i \geq sgn(j)(R_j - R_0), j = -s, \ldots, l & \\ & \lambda_i, \lambda_c \geq 0. & \end{array} \tag{2.13}$$

3 Solution and its properties

Although there are many software packages for solving linear programming problems, they cannot provide insights into the nature of the solution and its financial meaning. In this section we derive an explicit solution for the linear programming problem in the previous section by working on the primal and dual problem simultaneously. We will also point out the financial meaning of the solutions when appropriate.

3.1 Simplifying the dual problem

The structure of the problem allows some simplifications of the dual problem. Separating the constraints corresponding to y_j according to the sign leads to

$$u = \min_{(-j)=-1,\ldots,-s} \frac{R_{(-j)} - R_0}{I_{(-j)}} \geq \lambda_i \geq \max_{j=1,\ldots,l} \frac{R_j - R_0}{I_j} = l. \tag{3.1}$$

Given λ_i the constraints corresponding to x_k tell us that

$$\lambda_c \geq \max_{k=1,\ldots,n} \frac{r_k - R_0 - \lambda_i i_k}{c_k} \tag{3.2}$$

Suppose asset corresponding to k_1 is involved in the optimal balance sheet, then $x_{k_1} > 0$. This means the constraint corresponding to x_{k_1} in the dual problem must be binding. Inequality (3.2) tells us that we must have

$$\lambda_c = \frac{r_{k_1} - R_0 - \lambda_i i_{k_1}}{c_{k_1}} = \max_{k=1,\ldots,n} \frac{r_k - R_0 - \lambda_i i_k}{c_k}. \tag{3.3}$$

Since $\lambda_c \geq 0$ the dual value associated with (λ_i, λ_c) can be represented as

$$\begin{array}{rcl} c\lambda_c + i\lambda_i & = & c \max\left(\max_{k=1,\ldots,n} \frac{r_k - R_0 - \lambda_i i_k}{c_k}, 0\right) + i\lambda_i \\ & = & \max_{k=1,\ldots,n} \max\left(\frac{c(r_k - R_0)}{c_k} + \frac{ic_k - ci_k}{c_k}\lambda_i, i\lambda_i\right). \end{array} \tag{3.4}$$

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Defining linear functions

$$L_k(\lambda_i) = \frac{c(r_k - R_0)}{c_k} + \frac{ic_k - ci_k}{c_k}\lambda_i, \tag{3.5}$$

we can rewrite the dual problem as

$$d = \min_{\lambda_i \in [\max(l,0),u]} \max_{k=1,\ldots,n} \max(L_k(\lambda_i), i\lambda_i). \tag{3.6}$$

Remark 3.1. *When* $(r_k - R_0)/i_k \geq u$, $\max(L_k(\lambda_i), i\lambda_i) = L_k(\lambda_i)$ *on* $[\max(l,0), u]$.

3.2 Bonds

From inequalities (3.1) we can see that in order for the dual constaints to be feasible (primal solution bounded) we must have

$$\min_{j=1,\ldots,s} \frac{R_{-j} - R_0}{I_{-j}} \geq \max_{j=1,\ldots,l} \frac{R_j - R_0}{I_j}. \tag{3.7}$$

This is a condition ensuring that the bonds have no interest rate risk arbitrage. In fact, if this condition is violated, then we can find j_1 and j_2 such that

$$\frac{R_{-j_2} - R_0}{I_{-j_2}} < \frac{R_{j_1} - R_0}{I_{j_1}}.$$

Choose $y_{j_1}, y_{-j_2} > 0$ such that $y_{j_1} I_{j_1} = y_{-j_2} I_{-j_2}$. Then the balance sheet y_{j_1}, y_{-j_2} is interest risk neutral; yet it carries a positive return

$$(R_{j_1} - R_0)y_{j_1} - (R_{-j_2} - R_0)y_{-j_2} = y_{-j_2} I_{-j_2} \left(\frac{R_{j_1} - R_0}{I_{j_1}} - \frac{R_{-j_2} - R_0}{I_{-j_2}} \right) > 0.$$

Such an interest risk arbitrage often happens to CD which usually has an interest rate that is lower than that of the Central bank rate so that we will even have $(R_{-j_2} - R_0)/I_{-j_2} < 0$. However, the total available CD funds are often limited. Thus, in case that there are multiple bonds satisfying the inequality to take full advantage of such an interest risk arbitrage, one should choose the bond $j_1 = j^*$ such that

$$\frac{R_{j^*} - R_0}{I_{j^*}} = \max_{j=1,\ldots,l} \frac{R_j - R_0}{I_j}. \tag{3.8}$$

We summarized the results below.

Proposition 3.2. (Interest rate risk arbitrage) *If there exists* $(-j) \in \{-s, \ldots, -1\}$ *such that*

$$\frac{R_{(-j)} - R_0}{I_{(-j)}} < \max_{j=1,\ldots,l} \frac{R_j - R_0}{I_j}.$$

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Then there is an interest risk arbitrage involving short bond $(-j)$. For a fixed amount $y_{(-j)}$ of bond $(-j)$, define j^* as in (3.8); then the balance sheet $y_{(-j)}, y_{j^*} = y_{(-j)}I_{(-j)}/I_{j^*}$ and $y_0 = y_{j^*} - y_{(-j)}$ is the best to take advantage of this interest risk arbitrage with an expected return of

$$y_{(-j)}I_{(-j)}\left(\frac{R_{j^*} - R_0}{I_{j^*}} - \frac{R_{(-j)} - R_0}{I_{(-j)}}\right).$$

Next we turn to relations between quantities $0, l, u$.

Proposition 3.3. *If*

$$l = \max_{j=1,\ldots,l} \frac{R_j - R_0}{I_j} < 0,$$

then the optimal balance sheet cannot contain any long bond. Thus, a necessary condition for a long bond to involved in the optimal balance sheet is

$$l = \max_{j=1,\ldots,l} \frac{R_j - R_0}{I_j} > 0. \tag{3.9}$$

Proof. This is because in this case the constraints in the dual problem corresponding to y_j will be non-binding for all j so that $y_j = 0, j = 1, \ldots, l$. If

$$l = \max_{j=1,\ldots,l} \frac{R_j - R_0}{I_j} = 0,$$

then the long bond has the same return as the riskless Central bank fund that funds its holding and it is certainly not worth holding in the balance sheet. Q.E.D.

The financial meaning of Proposition 3.3 is clear: involving long bonds in a balance sheet makes sense only when they have a yield that is higher than the riskless Central bank fund rate which will fund them.

Also, we must have

$$u \geq 0. \tag{3.10}$$

That is, R_0 cannot be greater than any of the $R_{(-j)}$. In other words the model does not apply when the yield curve is inverted or there are liabilities such as CD which requires a lower interest payment than the Central bank fund. However, this is a technical requirement related to our conversion of the original problem. When such situation arises we need only to designate the liabilities with the lowest interest payment with index $j = 0$. We will elaborate this case later in Section 2.3.4. In the discussion below we will focus on the most common case when both conditions (3.9) and (3.10) hold.

We start our discussion from the simple situation that the only available assets and liabilities are bonds that do not have credit risk. Let $j^* \in \{1, \ldots, l\}$ be as defined in (3.8) and similarly let $(-j)^* \in \{-s, \ldots, -1\}$ be the indices such that

$$(R_{(-j)^*} - R_0)/I_{(-j)^*} = \min_{-j=-s,\ldots,-1} \frac{R_{-j} - R_0}{I_{-j}}. \tag{3.11}$$

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If multiple indices achieve the maximum or minimum, then the effect of the corresponding bonds in the balance sheet will be equivalent. For the simplicity of the analysis let's assume for now that the indices j^* and $(-j)^*$ are unique. These are the only bonds possibly involved in an optimal balance sheet in the primal problem. It is then natural to ask which one is actually involved. The next proposition provides the answer:

Proposition 3.4. *Suppose that no general asset is involved in the balance sheet and there does not exist interest rate risk arbitrage. Moreover, assume that in equality (3.9) holds. Then the optimal balance sheet contains a $y_{j^*} = i/I_{j^*}$ share of long bond j^*. It is funded by the same quantity of Central bank borrowing $y_0 = y_{j^*}$.*

Proof. If there is no general asset then the cost function of the dual problem is $i\lambda_i$. Therefore, the dual optimal solution $\lambda_i^* = (R_{j^*} - R_0)/I_{j^*} > 0$ by (3.9). Thus, the primal interest constraint is binding. Since the only binding dual constraint is the one corresponding to y_{j^*}, the optimal balance sheet contains only bond j^* besides the Central bank funding. It follows that the optimal balance sheet contains a $y_{k^*} = i/I_{k^*}$ share of long bond j^*. It is funded by the same quantity of Central bank borrowing $y_0 = y_{k^*}$. Q.E.D.

Remark 3.5. *The bank is using the Central bank funds to finance a long bond so as to make the return on interest difference. Of course, the interest difference must be positive implying the yield curve is not inverted. Proposition 3.4 says that one should chose the long bond with the maximum return to interest risk ratio so as to gain maximum return under the interest risk constraint.*

3.3 Optimal balance sheet involving general assets

An important and interesting question to ask is under what circumstances it is worth to involve a general asset in the optimal balance sheet. We have observed in Subsection 2.3.1 that if a general asset k_1 is involved in the optimal balance sheet then equality (3.3) holds. We know that $\lambda_c \geq 0$, hence the following characterization:

Proposition 3.6. *The optimal balance sheet contains at least one general asset if and only if*

$$\max_{k=1,\dots,n} \frac{r_k - R_0}{i_k} \geq \max(l, 0). \tag{3.12}$$

Remark 3.7. *Financially, Proposition 3.6 is quite intuitive. It says that in order for the optimal balance sheet to involve a general asset the maximum of the interest rate adjusted returns of all general assets has to be at least no less than that of the long bonds and nonnegative.*

Assuming condition (3.12) holds, we further discuss the construction of the optimal balance sheet. Recall that the dual problem can be represented as (3.6). We note that

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the function

$$\phi(\lambda_i) = \max_{k=1,\dots,n} \max(L_k(\lambda_i), i\lambda_i) \tag{3.13}$$

as the maximum of convex functions is a convex function. The dual problem is then

$$d = \min_{\lambda_i \in [\max(l,0),u]} \phi(\lambda_i). \tag{3.14}$$

Proposition 3.8. *Suppose that function ϕ is monotone decreasing on $[\max(l,0),u]$ (see Fig 1) we select k^* such that*

$$L_{k^*}(u) = \max_{k=1,\dots,n} L_k(u),$$

and consider two cases:

(a) *if*

$$\frac{r_{k^*} - R_0}{i_{k^*}} > \frac{R_{(-j)^*} - R_0}{I_{(-j)^*}}, \tag{3.15}$$

the optimal balance sheet is $x_{k^*} = c/c_{k^*}$, $y_{(-j)^*} = (i_{k^*}c - c_{k^*}i)/c_{k^*}I_{(-j)^*}$ and $y_0 = x_{k^*} - y_{(-j)^*}$, The corresponding dual solution is

$$(\lambda_i^*, \lambda_c^*) = \left(\frac{R_{(-j)^*} - R_0}{I_{(-j)^*}}, \left(\frac{r_{k^*} - R_0}{i_{k^*}} - \frac{R_{(-j)^*} - R_0}{I_{(-j)^*}} \right) \frac{i_{k^*}}{c_{k^*}} \right).$$

(b) *otherwise the optimal balance sheet is $x_{k^*} = i/i_{k^*}$, and $y_0 = x_{k^*}$ with dual solution*

$$(\lambda_i^*, \lambda_c^*) = \left(\frac{R_{(-j)^*} - R_0}{I_{(-j)^*}}, 0 \right).$$

Proof. Let $(\lambda_i^*, \lambda_c^*)$ be the optimal dual solution. Since ϕ is decreasing, the minimum of ϕ is attained at u. That is to say,

$$\lambda_i^* = (R_{(-j)^*} - R_0)/I_{(-j)^*} = u > 0. \tag{3.16}$$

It follows from the complementary slackness condition that the primal constraint for the interest risk must be binding. Moreover, the only dual constrain that is possibly binding is the one in (3.16). Thus, only a short bond $(-j)^*$ is possibly involved in the optimal balance sheet (if there are multiple $-j$ satisfying (3.16) then the corresponding short bonds have the same effect). Similarly, the only general asset that is possibly involved in the optimal balance sheet is k^* from its definition.

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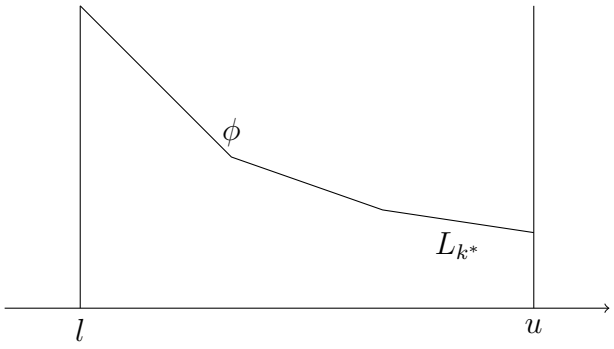


Figure 1: Decreasing ϕ.

We turn to the construction of the optimal balance sheet. We note that in this case $\phi(\lambda_i)$ cannot be $i\lambda_i$ as λ_i approaches u and, therefore, must coincide with L_{k^*}. This implies that $(r_{k^*} - R_0)/i_{k^*} > u$. From equality

$$d = i\lambda_i^* + c\lambda_c^* = L_{k^*}(\lambda_i^*), \tag{3.17}$$

we have

$$\lambda_c^* = \frac{r_{k^*} - R_0 - \lambda_i^* i_{k^*}}{c_{k^*}} = \left(\frac{r_{k^*} - R_0}{i_{k^*}} - \frac{R_{(-j)^*} - R_0}{I_{(-j)^*}}\right)\frac{i_{k^*}}{c_{k^*}}. \tag{3.18}$$

If in addition condition (3.15) is satisfied then we also have $\lambda_c^* > 0$ so that the primal constraint on credit risk is also binding. Solving the primal constraint we derive the optimal balance sheet $x_{k^*} = c/c_{k^*}$, $y_{(-j)^*} = (i_{k^*}c - c_{k^*}i)/c_{k^*}I_{(-j)^*}$ and $y_0 = x_{k^*} - y_{(-j)^*}$. If condition (3.15) fail then involve short bond $(-j)^*$ to hedge the interest risk will not increase the total balance sheet return. Thus, short bond $(-j)^*$ should not be in the optimal balance sheet. It follows that the optimal balance sheet must be $x_{k^*} = i/i_{k^*}$, and $y_0 = x_{k^*}$. Q.E.D.

Remark 3.9. *Note that function ϕ attains minimum at $\lambda_i^* = u$ implies that*

$$\frac{i}{i_{k^*}} \leq \frac{c}{c_{k^*}}. \tag{3.19}$$

Inequalities (3.15) and (3.19) characterize the environment in which the optimal balance sheet involves only general asset k^* and with a short bond together with the Central bank borrowing. Condition (3.19) states that given the credit risk budget c and interest rate risk budget i, when investing in the optimal asset, the interest rate risk will be exhausted first. Therefore, in order to consume all credit risk, the bank will need to hedge the interest rate risk with the long liability with the least interest rate risk cost. Condition (3.15) states that the interest rate risk-adjusted return on asset k^* should be higher than the interest

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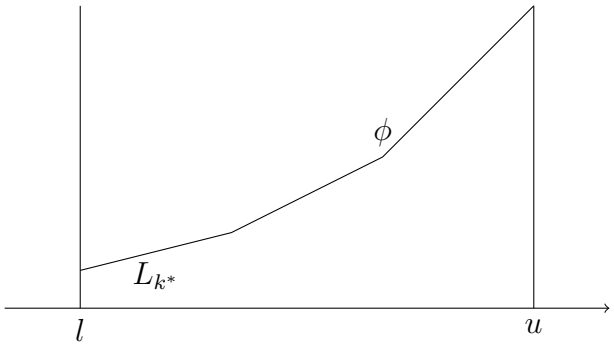


Figure 2: Increasing ϕ.

rate risk-adjusted cost on the liability $(-j)^*$. We note that in such a balance sheet, the interest risk of asset k^* in general exceeds the interest rate risk limit i, and sometimes by a lot.

On the other hand if (3.15) is not satisfied then there is no reason to hedge the interest risk using short bond $(-j)^*$. In this case all the interest risk is taken by the general asset k^* and there is may have a surplus in credit risk.

Similarly we can derive that the environment in which the optimal balance sheet involves only general asset k^* possibly together with a long bond both funded with the Central bank borrowing.

Proposition 3.10. *Suppose that function ϕ is monotone increasing on $[\max(l,0),u]$ (see Fig 2) we select k^* such that*

$$L_{k^*}(\max(l,0)) = \max_{k=1,\dots,n} L_k(\max(l,0))$$

If $l \le 0$ then the optimal balance sheet is $x_{k^*} = c/c_{k^*}$ and $y_0 = x_{k^*}$ and the dual solution is

$$(\lambda_i^*, \lambda_c^*) = \left(0, \frac{r_{k^*} - R_0}{c_{k^*}}\right);$$

otherwise we consider two cases

(a) *if*

$$\frac{r_{k^*} - R_0}{i_{k^*}} > \frac{R_{j^*} - R_0}{I_{j^*}}. \tag{3.20}$$

then the optimal balance sheet is $x_{k^*} = c/c_{k^*}$, $y_{j^*} = (c_{k^*} i - i_{k^*} c)/c_{k^*} I_{j^*}$ and $y_0 = x_{k^*} + y_{j^*}$, and the corresponding dual solution is

$$(\lambda_i^*, \lambda_c^*) = \left(\frac{R_{j^*} - R_0}{I_{j^*}}, \left(\frac{r_{k^*} - R_0}{i_{k^*}} - \frac{R_{j^*} - R_0}{I_{j^*}}\right)\frac{i_{k^*}}{c_{k^*}}\right).$$

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(b) *otherwise the optimal balance sheet is* $y_{j^*} = i/I_{j^*}$ *and* $y_0 = y_{j^*}$ *with dual solution*

$$(\lambda_i^*, \lambda_c^*) = \left(\frac{R_{j^*} - R_0}{I_{j^*}}, 0 \right).$$

Proof. Again, let $(\lambda_i^*, \lambda_c^*)$ be the optimal dual solution. When $l \leq 0$, no long bond is involved in the optimal balance sheet. Moreover, $\lambda_i^* = 0$ so that by (3.17)

$$\lambda_c^* = \frac{r_{k^*} - R_0}{c_{k^*}} > 0.$$

This means that the primal constraint corresponding to the credit risk is binding. Thus, the optimal balance sheet must be $x_{k^*} = c/c_{k^*}$ and $y_0 = x_{k^*}$.

If $l > 0$ then $\lambda_i^* = l > 0$ so that the primal constraint corresponding to the interest risk is binding. If (3.20) holds then by

$$\lambda_c^* = \frac{r_{k^*} - R_0 - \lambda_i^* i_{k^*}}{c_{k^*}} = \left(\frac{r_{k^*} - R_0}{i_{k^*}} - \frac{R_{j^*} - R_0}{I_{j^*}} \right) \frac{i_{k^*}}{c_{k^*}}. \tag{3.21}$$

we also have $\lambda_c^* > 0$, which implies that the primal constraint corresponding to the credit risk is also binding. Solving the two binding primal constraints we derive the optimal balance sheet $x_{k^*} = c/c_{k^*}$, $y_{j^*} = (c_{k^*} i - i_{k^*} c)/c_{k^*} I_{j^*}$ and $y_0 = x_{k^*} + y_{j^*}$. Otherwise, if (3.20) is false then using the interest risk on the long bond j^* is more favorable. Thus, the optimal balance sheet will be $y_{j^*} = i/I_{j^*}$ and $y_0 = y_{j^*}$. Q.E.D.

Remark 3.11. *When* ϕ *attains a minimum at*

$$\lambda_i^* = \max(0, (R_{j^*} - R_0)/I_{j^*}), \tag{3.22}$$

we have

$$\frac{i}{i_{k^*}} \geq \frac{c}{c_{k^*}}. \tag{3.23}$$

The nontrivial case is when $\lambda_i^* = (R_{j^*} - R_0)/I_{j^*} > 0$ *and the primal constraint corresponding to the interest risk is binding. Besides the Central Bank funding, the only possible assets to involve in the optimal balance sheet are general asset* k^* *and the long bond* j^**. If (3.20) holds then the general asset* k^* *takes priority, which will exhaust the credit risk first with* $x_{k^*} = c/c_{k^*}$*, and due to inequality (3.23) there is a possible surplus in interest risk which will be filled with the long bond* j^**. However, if (3.20) fails then the long bond* j^* *should take priority for the interest rate risk budget. In fact, in this case there is no need to involve the general asset* k^* *in the optimal balance sheet because its interest risk adjusted return is lower than that of the long bond* j^**, while in addition it also has credit risk exposure.*

The case when the minimum of function ϕ is attained inside $(\max(u, 0), l)$ (see Fig. 3) is more complicated.

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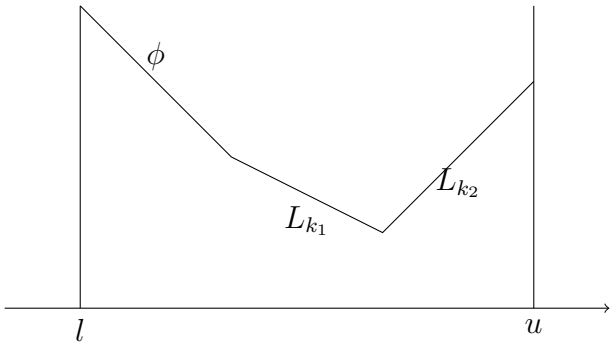


Figure 3: ϕ with minimum inside (l, u).

Proposition 3.12. *Suppose that function ϕ attains the minimum inside $(\max(l, 0), u)$; then there are two posible cases:*

(a) *function ϕ attains the minimum at the intersection of the graphs of L_{k_1} and L_{k_2}. Then $(ic_{k_1} - ci_{k_1})(ic_{k_2} - ci_{k_2}) < 0$ and the optimal balance sheet is $y_0 = x_{k_1} + x_{k_2}$, and*

$$(x_{k_1}, x_{k_2}) = \left(\frac{\begin{vmatrix} i & i_{k_2} \\ c & c_{k_2} \end{vmatrix}}{\begin{vmatrix} i_{k_1} & i_{k_2} \\ c_{k_1} & c_{k_2} \end{vmatrix}}, \frac{\begin{vmatrix} i_{k_1} & i \\ c_{k_1} & c \end{vmatrix}}{\begin{vmatrix} i_{k_1} & i_{k_2} \\ c_{k_1} & c_{k_2} \end{vmatrix}} \right). \tag{3.24}$$

The dual solution is

$$(\lambda_i^*, \lambda_c^*) = \left(\frac{\begin{vmatrix} c_{k_1} & r_{k_1} - R_0 \\ c_{k_2} & r_{k_2} - R_0 \end{vmatrix}}{\begin{vmatrix} c_{k_1} & i_{k_1} \\ c_{k_2} & i_{k_2} \end{vmatrix}}, \frac{\begin{vmatrix} r_{k_1} - R_0 & i_{k_1} \\ r_{k_2} - R_0 & i_{k_2} \end{vmatrix}}{\begin{vmatrix} c_{k_1} & i_{k_1} \\ c_{k_2} & i_{k_2} \end{vmatrix}} \right) \tag{3.25}$$

and the common primal and dual value is

$$p = d = \frac{\begin{vmatrix} ic_{k_1} - ci_{k_1} & r_{k_1} - R_0 \\ ic_{k_2} - ci_{k_2} & r_{k_2} - R_0 \end{vmatrix}}{\begin{vmatrix} c_{k_1} & i_{k_1} \\ c_{k_2} & i_{k_2} \end{vmatrix}}. \tag{3.26}$$

(b) *function ϕ attains the minimum at the intersection of the graphs of L_{k^*} and $i\lambda_i$. Then $(ic_{k^*} - ci_{k^*}) < 0$ and the optimal balance sheet is $y_0 = x_{k^*} = i/i_{k^*}$. The dual solution is*

$$(\lambda_i^*, \lambda_c^*) = ((r_{k^*} - R_0)/i_{k^*}, 0)$$

and the common primal and dual value is $p = d = i(r_{k^*} - R_0)/i_{k^*}$.

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Proof. When ϕ attains the minimum inside $(\max(l,0),u)$ at λ_i we must have $0 \in \partial\phi(\lambda_i)$ which means λ_i must be at the intersection of either (a) two lines L_{k_1} and L_{k_2} such that $(ic_{k_1} - ci_{k_1})(ic_{k_2} - ci_{k_2}) < 0$ or (b) L_{k^*} and $i\lambda_i$ such that $(ic_{k^*} - ci_{k^*}) < 0$. For case (a),

$$\lambda_c = \frac{r_{k_1} - R_0 - \lambda_i i_{k_1}}{c_{k_1}} = \frac{r_{k_2} - R_0 - \lambda_i i_{k_2}}{c_{k_2}} = \max_{k=1,\ldots,n} \frac{r_k - R_0 - \lambda_i i_k}{c_k}. \tag{3.27}$$

We can rewrite (3.27) as

$$\begin{aligned} c_{k_1}\lambda_c + i_{k_1}\lambda_i &= r_{k_1} - R_0 \\ c_{k_2}\lambda_c + i_{k_2}\lambda_i &= r_{k_2} - R_0. \end{aligned} \tag{3.28}$$

Solving (3.28), we derive the dual solution (3.25). Since in this case we must have $\lambda_i \in (l,u)$ and $\lambda_c > 0$ both primal constraints are binding and the optimal balance sheet does not contain any bonds. It follows that

$$\begin{aligned} i_{k_1}x_{k_1} + i_{k_2}x_{k_2} &= i \\ c_{k_1}x_{k_1} + c_{k_2}x_{k_2} &= c. \end{aligned} \tag{3.29}$$

Solving (3.29) we derive (3.24). Calculating the primal and dual value in (3.26) is straightforward.

In case (b), since there are no bonds involved in the optimal balance sheet and $\lambda_i^* > 0$, the primal constraint on interest rate risk is binding. Thus, $y_0 = x_{k^*} = i/i_{k^*}$. The calculation for the dual solution and the optimal value is straightforward.

Q.E.D.

Remark 3.13. *Geometrically, in the λ_i-return space, each general asset is represented by a line L_k. The boundary of the constraints on λ_i is represented by vertical lines of the form $\lambda_i = (R_j - R_0)/I_j$ and $\lambda_i = (R_{-j} - R_0)/I_{-j}$ corresponding to possible long and short bonds to be involved in the optimal balance sheet, respectively. Moreover $\lambda_c \geq 0$ is represented by the line $i\lambda_i$. Function ϕ is the maximum of the graphs of all these lines.*

Propositions 3.8, 3.10 and 3.12 tell us that, in general, the dual solution λ_i^* is attained at the intersection of two of the lines mentioned above. In general, the solution is a pair of assets complementary in their exposure to interest and credit risks, respectively. When the intersection occurs between two lines L_{k_1} and L_{k_2}, the optimal balance sheet is a pair of complementary general assets. When the intersection is a line L_k with a vertical line then the optimal balance sheet is the pair of the general asset k and either a long bond or a short bond. The conditions characterizes what kind of pair will occur and their financial meaning were explained in Remarks 3.9, 3.11 and the proof of Proposition 3.12. The exceptional case is when the optimal balance sheet corresponding to the intersection of some L_{k^*} and $i\lambda_i$. In this case, only one general asset k^* is in the optimal balance sheet. This feature of the optimal balance sheet is very useful in further discussion of practical optimal balance sheets involving diversification constraints in Section 7 below.

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Of course, it is possible that the dual solution λ_i^* is attained at the intersection of three or more lines related to financial assets. However, in this case at most two suffice in generating the optimal balance sheet. Adding more will not improve the return of the optimal balance sheet.

We note that Propositions 3.8, 3.10 and 3.12 do not provide explicit procedure for finding the optimal balance sheet. We will pick up that topic in the sequel.

3.4 Inverted yield curve

If the interest rate on Central bank funds is not the lowest among all the interest rates for liabilities then we can designate the index $j = 0$ to the one with the lowest interest. The conclusions in the previous sections remain valid after the small and obvious modification of replacing

$$\frac{r_{k^*} - R_0}{i_{k^*}}$$

by

$$\frac{r_{k^*} - R_0}{i_{k^*} - I_0},$$

and

$$\frac{R_{j^*} - R_0}{I_{j^*}} \text{ and } \frac{R_{(-j)^*} - R_0}{I_{(-j)^*}}$$

by

$$\frac{R_{j^*} - R_0}{I_{j^*} - I_0} \text{ and } \frac{R_{(-j)^*} - R_0}{I_{(-j)^*} - I_0}$$

respectively.

3.5 Finding the solution

The discussion in the previous sections suggests the following procedure of solving the linear programming problems related to the risk allocation problem. To be concise we consider the case when the Central bank funding has the lowest interest rate among all funding sources. When this is not true we can use the modification described in Section 2.3.4. We continue to use the notation $j^* \in \{1, \ldots, l\}$ and $(-j)^* \in \{-s, \ldots, -1\}$ as defined in (3.8) and (3.16), respectively.

1. If $(R_{j^*} - R_0)/I_{j^*} > (R_{(-j)^*} - R_0)/I_{(-j)^*}$ then there is an interest risk arbitrage as described in Proposition 3.2.

2. If $(R_{j^*} - R_0)/I_{j^*} < 0$ then no long bond is involved in the optimal balance sheet. This corresponds to an inverted yield curve, so that the interest rate on Central bank funding is higher than that of any of the longer term bonds. We will assume this does not happen below so that $l > 0$.

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3. If there are no general assets in the balance sheet then the long bond j^* is involved in the optimal balance sheet besides the Central bank funding as described in Proposition 3.4.

4. We turn to the case when general assets are involved. For convenience we denote $L_0(\lambda_i) = i\lambda_i$. Choose k_1 and k_2 such that $L_{k_1}(l) = \max_k L_k(l)$, and $L_{k_2}(u) = \max_k L_k(u)$, respectively.

5. If both L_{k_1} and L_{k_2} are nonincreasing, then only asset $k^* = k_2$ is possibly involved in the optimal balance sheet which is determined by Proposition 3.8.

6. If both L_{k_1} and L_{k_2} are nondecreasing, then only asset $k^* = k_1$ is possibly involved in the optimal balance sheet which is determined by Proposition 3.10.

7. It remains to examine the case when L_{k_1} is decreasing and L_{k_2} is increasing. Define

$$m = \begin{cases} (r_{k_1} - R_0)/i_{k_1}, & \text{if} \quad k_2 = 0; \\ \dfrac{\begin{vmatrix} c_{k_1} & r_{k_1} - R_0 \\ c_{k_2} & r_{k_2} - R_0 \end{vmatrix}}{\begin{vmatrix} c_{k_1} & i_{k_1} \\ c_{k_2} & i_{k_2} \end{vmatrix}}, & \text{otherwise.} \end{cases}$$

8. Select $\bar{k}$ such that $v_m = L_{\bar{k}}(m) = \max_k L_k(m)$. First consider the case when

$$v_m > L_{k_1}(m) = L_{k_2}(m).$$

   If $ic_{\bar{k}} - ci_{\bar{k}} < 0$ set $k_1 = \bar{k}$; otherwise set $k_2 = \bar{k}$ then go back to Step [7.]

9. If $v_m = L_{k_1}(m) = L_{k_2}(m)$, then the optimal balance sheet is determined by Proposition 3.12.

The cases when there are no general assets involved are straight forward. We discuss Steps 4–9. After finding out the indices k_1 and k_2 corresponding to assets that attain a maximum at l and u, respectively, we need to find the minimum point of the function ϕ. If both L_{k_1} and L_{k_2} are nonincreasing as in Step 5, then no part of the line segment

$$L_k(t),\ t \in [l, u]$$

with L_k nonincreasing can go above the two lines indexed by k_1 and k_2. Thus, ϕ must be a decreasing function so that it attains minimum at the right end point u. Similarly, the argument applies to the case discussed in Step 6. The loop described in Step 7 and Step 8 will lead to Step 9 in at most $n-2$ iterations. The equality in Step 9 implies that

$$0 \in \partial\phi(m),$$

because m is the point where the decreasing linear function L_{k_1} and the increasing linear function L_{k_2} intersects. Thus, m is the minimum of ϕ inside the interval (l, u) and the optimal balance sheet can be determined by Proposition 3.12.

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Remark 3.14. *In the cases corresponding to Steps 5 and 6, the optimal balance sheet needs only one general asset and in the case corresponding to Step 9 the optimal balance sheet needs at most two general assets. There could be additional general assets that meets the same criteria, however, involving them will not change the return of the optimal balance sheet.*

4 Dual solution as the prices for buying insurance on the risks

Financially, the dual solution can be explained as the prices to buy the corresponding insurance for interest rate and credit risks under the assumption of no arbitrage. First we make precise what we mean by arbitrage.

Definition 4.1. *Let $\bar{\lambda}_i$, and $\bar{\lambda}_c$ be the price for buying insurance on one unit of the interest risk and credit risk, respectively. We say they provide an arbitrage opportunity provided that the bank can either (a) construct a riskless balance sheet that makes a positive return or (b) achieve a better risk adjusted return as one would holding a balance sheet but without doing so.*

Item (a) in the definition is standard. We need (b) is because our balance sheet cannot be shorted. Condition (b) is reasonable in that constructing balance sheets is a core bank operation. If one can replace it with a simple trade that amounts a 'free lunch'.

Theorem 4.2. *Let $\bar{\lambda}_i$, and $\bar{\lambda}_c$ be the price for buying insurance on one unit of interest rate risk and credit risk, respectively. Assume that there is no arbitrage as described in Definition 4.1. Then $(\bar{\lambda}_i, \bar{\lambda}_c)$ is a dual solution.*

Proof. First we show that under the assumption of no arbitrage, $\bar{\lambda}_i$ and $\bar{\lambda}_c$ satisfy the dual constraints. Suppose that $\bar{\lambda}_i$, $\bar{\lambda}_c$ violate the constraint corresponding to x_k so that

$$i_k\bar{\lambda}_i + c_k\bar{\lambda}_c < r_k - R_0. \tag{4.1}$$

Then we can establish a position x_k and buy insurance for the interest risk of x_k at a price $\bar{\lambda}_i x_k$ and buy the insurance for the credit risk of x_k at a price $\bar{\lambda}_c x_k$. This way the position is riskless and yet gains a positive return of

$$[(r_k - R_0) - i_k\bar{\lambda}_i - c_k\bar{\lambda}_c]x_k > 0.$$

This is an arbitrage. Similarly, suppose that $\bar{\lambda}_i$, $\bar{\lambda}_c$ violate the constraint corresponding to y_j. Without loss of generality let's consider the case when $j > 0$. Then

$$I_j\bar{\lambda}_j < R_j - R_0.$$

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In this case we can establish a position y_j and buy insurance for the interest rate risk of y_j at price $\bar{\lambda}_i y_j$; funding the buying with the Central bank borrowing creates a riskless position whose return is

$$(R_j - R_0 - I_j\bar{\lambda}_j)y_j > 0.$$

Again this is an arbitrage.

On the other hand if $i\bar{\lambda}_i + c\bar{\lambda}_c > d$ then we can sell i units of interest rate risk, c units of credit risk and do away with the operation of holding a primal optimal balance sheet. This way we assume the same risks i and c but by weak duality we will gain a higher return

$$i\bar{\lambda}_i + c\bar{\lambda}_c > d \geq p$$

than we would holding any feasible balance sheet. Thus, $\bar{\lambda}_i$ and $\bar{\lambda}_c$ must be the dual solution.

Q.E.D.

5 Examples and applications

In this section we apply our research to real data, taken from databases that report yields, interest rates, default rates and losses given defaults. By doing so, we show how our research is applied to a real problem.

We will use assets and liabilities described in Table 2 to illustrate various issues in bank balance sheet risk allocation problems. We assume that the upper limits of both interest and credit risks are 50, i.e. $i = c = 50$.

Returns for bonds and deposits are given by the respective yields at the end of 2018, which we take from the Federal Reserve of St. Louis (FRED) database. The interest rates for financial bonds are taken from the Bloomberg database.

Using the yields above, we can calculate the respective interest rate risk I_j, using the formulas (2.1) and (2.2).

For general assets, the returns are given by the interest rates, subtracted by the expected loss, which is equal to the product of the probability of default and the loss given default. Again, the interest rates were taken from the FRED database.

The probability of default was estimated by using the ten-year moving average of the default rate on each asset class. For mortgages and consumer credit, we backed the probability of default by using the charge-off rate, available in the FRED database. Corporate default rates were taken from Moody's (2018).

Losses given default for corporate bonds are also reported by Moody's (2018), whereas the data for mortgages was taken from Tung, Hu and Cantor (2006), and the data for consumer credit taken from Bandyopadhyay and Singh (2007).

The probabilities of default and LGDs allow us not only to estimate the expected returns for general assets, but also to estimate their credit risk, by resorting to formula (2.9).

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Asset class	allocation	r, R	returns	i, I	interest risk	c	credit risk
Mortgage 15 yrs	x_1	r_1	3.22%	i_1	10.35%	c_1	9.6%
Mortgage 30 yrs	x_2	r_2	3.76%	i_2	26.99%	c_2	9.9%
Corporate bonds BBB	x_3	r_3	5.02%	i_4	32.04%	c_3	1.6%
Consumer credit	x_4	r_4	7.68%	i_3	3.96%	c_4	14.3%
Corporate bonds AAA	y_1	R_1	4.02%	I_1	35.9%	N/A	N/A
Treasury 10 year	y_2	R_2	2.69%	I_2	19.96%	N/A	N/A
Treasury 5 year	y_3	R_3	2.51%	I_3	10.7%	N/A	N/A
Treasury 2 year	y_4	R_4	2.48%	I_4	4.44%	N/A	N/A
Liquidity	$-y_0$	R_0	2.4%	I_0	0%	N/A	N/A
Liability class	allocation	R	returns	I	interest risk		
Central bank funding	y_0	R_0	2.4%	I_0	0%	N/A	N/A
Comercial paper 1 month	y_{-1}	R_{-1}	2.42%	I_{-1}	0.19%	N/A	N/A
Comercial paper 3 months	y_{-2}	R_{-2}	2.67%	I_{-2}	0.57%	N/A	N/A
Financial bonds 2 year	y_{-3}	R_{-3}	3.29%	I_{-3}	4.39%	N/A	N/A
Financial bonds 5 year	y_{-4}	R_{-4}	3.53%	I_{-4}	10.39%	N/A	N/A
Financial bonds 10 year	y_{-5}	R_{-5}	3.93%	I_{-5}	18.75%	N/A	N/A
24 month deposit	y_{-6}	R_{-6}	0.81%	I_{-6}	4.55%	N/A	N/A
12 month deposit	y_{-7}	R_{-7}	0.61%	I_{-7}	2.29%	N/A	N/A

Table 2: Multi-asset example

We note that Liquidity can be viewed as the short of Central Bank Funding. When $y_0 < 0$ we should use liquidity instead. Usually the rate of the liquidity is slightly lower than that of the Central Bank Funding. Here we assume they are the same to simplify the exposition.

5.1 Interest risk arbitrage

We can see that $(R_{-7} - R_0)/I_{-7} = -0.78$ and $(R_{-6} - R_0)/I_{-6} = -0.34$ both are negative and smaller than $(R_1 - R_0)/I_1 = 0.045 = \max_{j=1,\ldots,5}(R_j - R_0)/I_j$, which indicate an interest rate risk arbitrage. In practice, however, we know that certificate of deposit funding is limited. Therefore, the practical implication is that the bank can construct such interest rate risk arbitrages as a stand alone position to the extent that the certificate of deposit funding is exhausted.

If we know that the the total available 12 and 24 month deposits are d_1 and d_2, respectively. Then by Proposition 3.2 the best balance sheets to take advantage of those CD are $y_{-7} = d_1, y_1 = y_{-7}I_{-7}/I_1$ and $y_0 = y_1 - y_{-7}$ and $y_{-6} = d_2, y_1 = y_{-6}I_{-6}/I_1$ and

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j	1	2	3	4
$(R_j - R_0)/I_j$	4.51%	1.45%	1.028%	1.8%

Table 3: Interest rate risk adjusted long bond returns

$-j$	-1	-2	-3	-4	-5
$(R_{(-j)} - R_0)/I_{(-j)}$	10.56%	47.67%	20.19%	10.87%	8.145%

Table 4: Interest rate risk adjusted short bond returns

$y_0 = y_1 - y_{-6}$, respectively. The total expected return of these two balance sheets are

$$d_1 I_{-7}\left(\frac{R_1 - R_0}{I_1} - \frac{R_{-7} - R_0}{I_{-7}}\right) + d_2 I_{-6}\left(\frac{R_1 - R_0}{I_1} - \frac{R_{-6} - R_0}{I_{-6}}\right).$$

Since these balance sheets are interest risk arbitrages they do not cause any interest rate risk. Thus, we can exclude the certificate of deposit funding from the balance sheet consideration.

5.2 Bonds only balance sheet

Consider the case that the only available assets and liabilities are bonds $j = -5, \ldots, 5$. Summarizing the ratio $(R_j - R_0)/I_j, j \in \{-5..5\}\backslash\{0\}$ in Tables 3 and 4 we can see that

$$\begin{aligned} & 0.08145 = (R_{-5} - R_0)/I_{-5} = \min_{(-j)}(R_{(-j)} - R_0)/I_{(-j)} \qquad (5.1) \\ > \quad & \max_j (R_j - R_0)/I_j = (R_1 - R_0)/I_1 = 0.0451 > 0. \end{aligned}$$

Then by Proposition 3.4 the optimal bond only balance sheet will contain only Corporate bonds AAA and Central bank funding. More concretely the optimal balance sheet is a $y_1 = i/I_1 = 139.28$ share of Corporate bonds AAA. It is funded by the same quantity of Central bank funding $y_0 = 139.28$. The total expected return of this balance sheet is $(R_1 - R_0)y_1 = 2.26$.

5.3 Balance sheet involving one general asset

We now consider the optimal balance sheet assuming that all the assets and liabilities in Table 2 are available except for consumer credit. As discussed in Subsection 5.1 we excludes the 12 and 24 month deposits so that $k \in \{1, 2, 3\}$, and $j \in \{-5, \ldots, 5\}$. Moreover, according to the discussion in Subsection 5.2 besides the Central bank funding, only Corporate bonds AAA that gives $l = (R_1 - R_0)/I_1 = 0.0451$ and the Financial

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k	0	1	2	3
L'_k	+	-	-	-
$L_k(l)$	2.26	4.09	2.97	38.08
$L_k(u)$	4.07	3.95	-0.145	4.35

Table 5: Determine general asset(s) in the optimal balance sheet

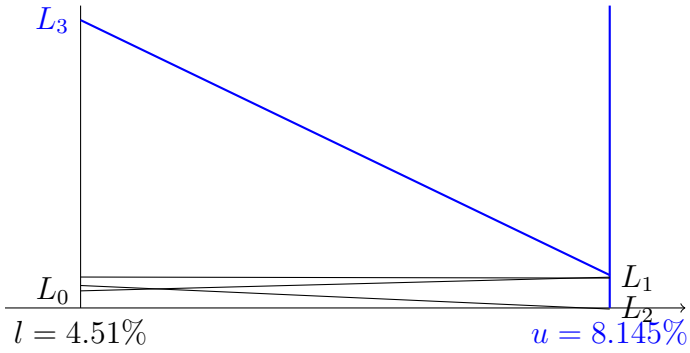


Figure 4: Graphic illustration for Table 5

bonds 10 year with $u = (R_{-5} - R_0)/I_{-5} = 0.08145$ will be involved in the balance sheet. According to Step 4 in Subsection 2.3.5, we compile Table 5.

From Table 5 we can read that $k_1 = k_2 = 3$ (see Fig. 4 for a graphic illustration). Thus, Corporate bonds BBB is the only general asset class in the optimal balance sheet. Moreover, $L'_3 \leq 0$ confirms that the condition in Step 5 of Subsection 3.5 is satisfied. Thus, by Proposition 3.8 Financial bond 10 year and the Central bank funding are the funding souces of the optimal balance sheet. The optimal balance sheet consists of

$$x_3 = \frac{c}{c_3} = 3040, y_{-5} = \frac{i_3}{I_{-5}}\left(\frac{c}{c_3} - \frac{i}{i_3}\right) = 4928, \text{ and } y_0 = x_3 - y_{-5} = -1888.$$

Its expected return from Table 5 is 4.35. By duality

$$50(\lambda_i^* + \lambda_c^*) = 4.35$$

By equation (3.8), we have $\lambda_i^* = 0.08145$. Then we can derive $\lambda_c^* = 0.0056$.

5.4 General solution

We now consider the optimal balance sheet assuming that all the assets and liabilities in Table 2 are available. This means adding consumer credit to the discussion in Subsection 5.3. Again as discussed in Subsection 5.1 we exclude the 12 and 24 month deposits.

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k	0	1	2	3	4
L_k'	+	-	-	-	+
$L_k(l)$	2.26	4.09	2.97	38.08	20.09
$L_k(u)$	4.07	3.95	-0.145	4.35	21.41

Table 6: Determine general asset(s) in the optimal balance sheet

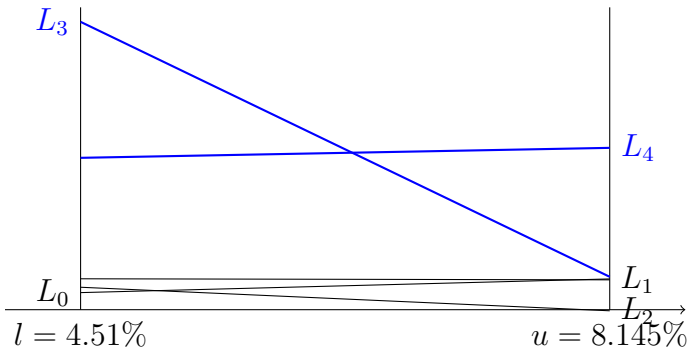


Figure 5: Graphic illustration for Table 6

Moreover, according to the discussion in Subsection 5.2, besides the Central bank funding, only Corporate bonds AAA with $(R_1 - R_0)/I_1 = 0.0451$ and the Financial bonds 10 year with $(R_{-5} - R_0)/I_{-5} = 0.08145$ are involved in the optimal balance sheet.

We expand Table 5 to Table 6. From Table 6 we can read that $k_1 = 3$ and $k_2 = 4$ (see graphic illustration in Fig. 5). Now that $L_3' < 0$ and $L_4' > 0$, we need to follow Steps 7-9 in Subsection 3.5.

First we calculate

$$m = \frac{\begin{vmatrix} c_3 & r_3 - R_0 \\ c_4 & r_4 - R_0 \end{vmatrix}}{\begin{vmatrix} c_3 & i_3 \\ c_4 & i_4 \end{vmatrix}} = 0.0638.$$

Then we tally the value of L_k at m in Table 7. We can see that $L_3(m) = L_4(m)$ shares the highest value 20.77, which is the expected return of the optimal balance sheet.

k	0	1	2	3	4
L_k'	+	-	-	-	+
$L_k(m)$	3.19	1.36	4.02	20.77	20.77

Table 7: Determine general asset(s) in the optimal balance sheet

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By Proposition 3.12 the optimal balance sheet consists of

$$(x_3, x_4) = \left(\frac{\begin{vmatrix} i & i_4 \\ c & c_4 \end{vmatrix}}{\begin{vmatrix} i_3 & i_4 \\ c_3 & c_4 \end{vmatrix}}, \frac{\begin{vmatrix} i_3 & i \\ c_3 & c \end{vmatrix}}{\begin{vmatrix} i_3 & i_4 \\ c_3 & c_4 \end{vmatrix}} \right) = (114, 336), \text{ and } y_0 = x_3 + x_4 = 450.$$

Its expected return is

$$(r_3 - R_0)x_3 + (r_4 - R_0)x_4 = 20.77,$$

verifying what we have seen in Table 7. It also follows from Proposition 3.12 that the dual solution is

$$(\lambda_i^*, \lambda_c^*) = \left(\frac{\begin{vmatrix} c_3 & r_3 - R_0 \\ c_4 & r_4 - R_0 \end{vmatrix}}{\begin{vmatrix} c_3 & i_3 \\ c_4 & i_4 \end{vmatrix}}, \frac{\begin{vmatrix} r_3 - R_0 & i_3 \\ r_4 - R_0 & i_4 \end{vmatrix}}{\begin{vmatrix} c_3 & i_3 \\ c_4 & i_4 \end{vmatrix}} \right) = (0.0638, 0.3515).$$

From the dual solution we can see that the marginal return on credit risk is much higher than that of the interest rate risk. This seems to suggest that the bank should allocate more of the total risk to credit risk. However, in reality consumer credit is of limited supply. Moreover, over concentration on one type of asset is unwanted.

Remark 5.1. *1. Compare the results in Subsections 5.3 and 5.4 we notice the dramatical flip of the sensitivity with respect to interest and credit risks.*

2. The interest arbitrage in Subsection 5.1 and huge advantage of holding consumers credit (incressing the expected return of the optimal balance sheet from 4.35 *to* 20.77*) show that the markets are far from efficient.*

6 Sensitivity and risk allocation

The dual solution can also be interpreted as the marginal rate of change of the expected return of the optimal balance sheet with respect to the change in the interest risk and credit risk limits. In fact, if the dual solution is $(\lambda_i^*, \lambda_c^*)$ strong duality implies that

$$p = d = i\lambda_i^* + c\lambda_c^*,$$

so that we can see that when we change i and c the marginal rate of change of the expected return of the optimal balance sheet is exactly λ_i^* and λ_c^*, respectively. Of course, this observation is heuristic. A rigorous argument showing that the dual solution is the subdifferential of the value of the primal problem as a function of the limits i and c on the interest and credit risk can be found in e.g. [Borwein & Zhu (2016)]. This view of the dual solution is rather useful in discussing how best to allocate the total risk to

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k	0	1	2	3
L'_k	+	-	-	-
$L_k(l)$	4.29	7.74	7.33	7.79
$L_k(u)$	7.74	4.47	4.36	7.79

Table 8: Determine the general asset(s) in the optimal balance sheet

interest rate and credit risks. The value of the dual solution is determined in Propositions 3.8, 3.10 and 3.12 along with optimal balance sheets.

We illustrate how this works using the examples in Subsection 5.3 and 5.4.

Example 6.1. *Let's first re-examine the example in Subsection 5.3. Assume that the total limit for risk is* $50 + 50 = 100$ *as in Subsection 5.3 but this limit for risk can be distributed to interest and credit risks, i.e., we assume* $i + c = 100$*. The dual solution* $(\lambda_i^*, \lambda_c^*) = (0.08145, 0.0056)$ *suggests to over weight the interest risk. However, it is clear that we cannot go 'all in' for the interest risk. That is because if* $i = 100$ *and* $c = 0$*, then there will be no general asset in the balance sheet. This amounts to reducing the problem to that of Subsection 5.2 which has optimal value* 2.26 *smaller than the optimal balance sheet in Subsection 5.3 has achieved.*

So the question is: to what extent we can overweight the interest risk limits? One way to think about it is that we need to increase i *while keeping the nonincreasing property of the linear function* L_3*. The most we can do given this property is make* $ic_3 - ci_3 = 0$ *which leads to* $i = 95.12$ *and* $c = 4.88$*. Using these values to re-calculate Table 5 we get Table 8. From Table 8 we can again read that* $k_1 = k_2 = 3$*. Thus, Corporate bonds BBB is still the only general asset class in the optimal balance sheet. Moreover,* $L'_3 \leq 0$ *confirms that the condition in Step 5 of Subsection 3.5 is satisfied. Thus, by Proposition 3.8 Financial bond 10 year and the Central bank funding are the funding souces of the optimal balance sheet. The optimal balance sheet consists of*

$$x_3 = \frac{c}{c_3} = 297, y_{-5} = \frac{i_3}{I_{-5}}\left(\frac{c}{c_3} - \frac{i}{i_3}\right) = 0, \text{ and } y_0 = x_3 = 297.$$

Its expected return from Table 8, however, is increased to 7.79*. Observe that as long as we have* $k_1 = k_2 = 3$ *by (b) of Proposition 3.8 the dual solution remains the same. Thus, changing the proportion of* i *and* c *in this range, the expected return changes exactly as the marginal rate of change that the dual solution indicates.*

Example 6.2. *Next we turn to the example in Subsection 5.4. Again assume that the total limit for risk can be distributed to interest rate and credit risks:* $i + c = 100$*. The dual solution* $(\lambda_i^*, \lambda_c^*) = (0.0638, 0.3515)$ *now suggests to overweight credit risk. Similar to Example 6.1 we seek the range in which the dual solution remains the same. The key is to maintain* L_3 *nonincreasing and* L_4 *nondecreasing. We note that when increasing* c*,*

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k	0	1	2	3	4
L'_k	+	-	-	-	0
$L_k(l)$	0.97	3.86	2.11	57.08	28.91
$L_k(u)$	1.76	1.57	-4.84	2.42	28.91

Table 9: Determine the general asset(s) in the optimal balance sheet

k	0	1	2	3	4
L'_k	0	-	-	-	0
$L_k(l)$	0	3.86	1.44	71.65	35.67
$L_k(u)$	0	-0.24	-8.44	0.84	34.66

Table 10: Determine general asset(s) in the optimal balance sheet

L_3 will always maintain the nonincreasing property. Thus we focus on L_4. Given that $i + c = 100$, solving $ic_4 = ci_4$ we can derive the upper bound $c = 78.3$ for L_4 to maintain the nondecreasing property. Indeed, we set $c = 78.3$ and $i = 21.7$ and update Table 6 to Table 9. From Table 9 we can see that as expected we still have $k_1 = 3$ and $k_2 = 4$. Thus, $m = 0.0638$ is unchanged. We can check out from Table 9 that the new expected return is increased to 28.91. The optimal balance sheet by Proposition 3.12 is $(x_3, x_4) = (0, 548)$.

Example 6.3. *In Example 6.2 limiting $c \leq 78.3$ is mainly for maintaining the dual solution unchanged. Besides that there is no reason that we cannot allocate the entire risk budget to the credit risk. So we try that by setting $c = 100$ and $i = 0$ and update Table 9 to Table 10: From Table 10 we can see that all the linear functions L_k are nonincreasing. Thus, ϕ is a nonincreasing function so that by Proposition 3.8 Consumer credit is the only general asset class in the optimal balance sheet. We further check that*

$$1.33 = \frac{r_4 - R_0}{i_4} > \frac{R_{-5} - R_0}{I_{-5}} = 0.081.$$

Hence by Proposition 3.8, the optimal balance sheet is

$$x_4 = \frac{c}{c_4} = 699, y_{-5} = \frac{i_4}{I_{-5}}\left(\frac{c}{c_4} - \frac{i}{i_4}\right) = 148, \text{ and } y_0 = x_4 - y_{-5} = 551.$$

Its expected return from Table 9 is increased to 34.66 from that of 28.91 in Example 6.2. However, the dual solution is changed in this case and the original dual solution in Subsection 5.4 cannot serve as a guide to trade-off the expected return with the balance of the interest rate and credit risks.

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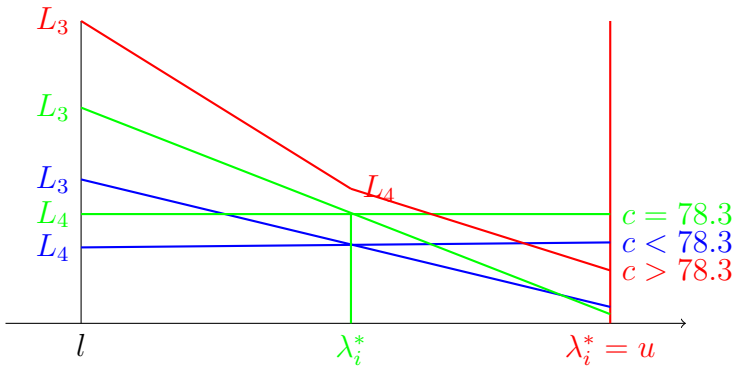


Figure 6: The impact of changing c in Examples 6.2 and 6.3.

Remark 6.4. *The dual solution tells us the marginal rate of change for interest and credit risks with respect to the distribution of their limits. This works to the extent that these changes happen in the range in which the dual solution remains unchanged. Within such a range the dual solution can be conveniently used to inform bank managers on how to trade-off between the expected return and the balance of the interest rate and credit risks. Outside such a range one may still be able to improve the expected return. However, the dual solution will change and, therefore, the marginal rates of change of the expected return with respect to the changes of interest and credit risks will also change. Fig 6 gives an illustration of the impact of changing the credit limit c in Examples 6.2 and 6.3. Note that when c is changing in the range* $c \leq 78.3$ *the dual solution* λ_i^* *does not change. However, when c increase beyound* 78.3, λ_i^* *will suddenly shift to the right endpoint of the domain u.*

7 Diversification constraints

Concentrating in at most two general asset and at most one bond in an optimal balance sheet is too risky to be practical. Good balance sheet management always places a cap on individual assets in a balance sheet. Nevertheless, the properties of the optimal balance sheet and the solutions discussed in Section 2 give us guidance on how to construct reasonable balance sheets under diversification constraints. One simple approach is combining properties of the optimal balance sheet with a greedy algorithm. We illustrate by reexamining the examples we have discussed above. Let us assume there is a cap that any individual asset cannot take more than 1/3 of the interest and credit risk limitations.

7.1 Bond only balance sheet

The simplest case is the bond only balance sheet. By Proposition 3.4 we need only to follow the decending order of the return to risk ratio $(R_j - R_0)I_j, j = 1, \ldots, l$ to fill the

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k	0	1	2
L'_k	+	-	-
$L_k(l)$	2.26	4.09	2.97
$L_k(u)$	5.28	3.85	-2.22

Table 11: Determine general asset(s) in the optimal balance sheet

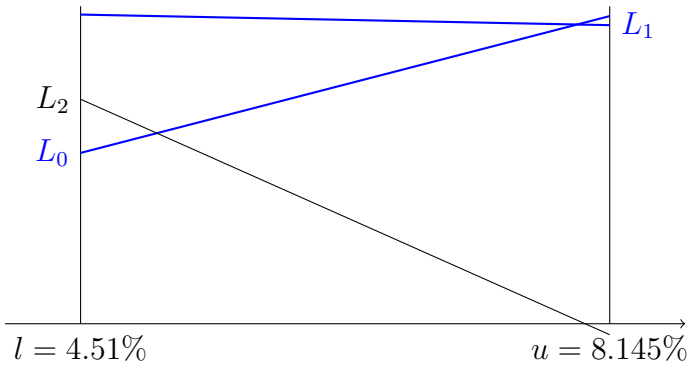


Figure 7: Graphic illustration for Table 11

cap for the interest risks. In our example we can compile Table 3. Thus the best balance sheet by combining Proposition 3.4 and a greedy algorithm is $y_1 = i/3I_1 = 46, y_2 = i/3I_2 = 84, y_4 = i/3I_4 = 375$ and $y_0 = y_1 + y_2 + y_4 = 505$.

7.2 Balance sheet involving one general asset

Step 1. Checking the solution in Subsection 6.3 we can see that the interest risk of the Corporate Bonds BBB $x_3 = 3040$ is $x_3 * i_3 = 3040 * 0.3204 = 974$, which far exceeds $i = 50$ itself. What happens here is that the interest risk is hedged by the funding source 10 Year Financial bonds $y_{-5} = 4928$ which have a negative interest rate risk of -924. From this observation, we see that we have to treat the pair of assets x_4 and y_{-5} as a compounded asset. Taking this point of view, using the principle of a greedy algorithm we should use 1/3 of this optimal pair at $x_4 = 1013$ and $y_{-5} = 1643$.

Step 2. We eliminate this pair and seek the next best solution. Without short bond -5 the next smallest value in Table 4 is 10.56%, corresponding the funding source commercial paper 1 month $-j = -1$. Thus, we need to update the upper bound of λ_i to $u = 10.56\%$. Updating Table 5 with this new value of u and eliminating column corresponding to $k = 3$, we get Table 11.

By Step 4 in Subsection 3.5 we have $k_1 = 1$ and $k_2 = 0$ (see Fig. 7 for a graphic illustration). By Step 7 of Subsection 3.5 we calculate that $m = (r_1 - R_0)/i_1 = 0.079$.

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k	0	1	2
L'_k	+	-	-
$L_k(m)$	3.96	3.96	0.062

Table 12: Determine general asset(s) in the optimal balance sheet

Calculating the value of L_k at m we get Table 12. We can see that L_0 and L_1 shares the highest value 3.96 at m. By case (a) of Proposition 3.12 the optimal balance sheet contains only the general asset 15 year Mortgage with $x_1 = i/3i_1 = 161$. Note that the credit risk of this posiiton is less than the limit $c/3$.

Step 3. Now disregard the 15 year Mortgage in Table 11; we are left with L_2 corresponding to the 30 year Mortgage and L_0. Examining the related columns in Table 11, by Step 4 in Subsection 3.5 we see that $k_2 = 0$ and $k_1 = 2$. Since in this case there are only two lines left, the condition in Step 9 of Subsection 3.5 must hold and we can determine the optimal balance sheet by case (b) in Proposition 3.12 as $x_2 = i/i_2 = 62$.

Step 4. The balance of central bank funding will be $y_0 = x_1 + x_2 + x_4 - y_{-5} = -407$.

7.3 General case

We now turn to the general case discussed in Subsection 6.4. Similarly to our discussion in 7.2, the pair of solution (x_3, x_4) in Subsection 6.4 should be viewed as an integrated group.

Step 1. Subject to the diversification constraints we need to scale them back to $(x_3, x_4) = (38, 112)$ and then eliminate them from the available assets.

Step 2. Eliminate assets 3 and 4; we get to a situation that has already been discussed in Subsection 7.2. Thus, the balance sheet contains only the general asset 15 year Mortgage with $x_1 = i/3i_1 = 161$.

Step 3. After further eliminating the 15 year Mortgage, we arrive at the same situation discussed in Step 3 in Section 7.2 and, therefore, we should include in the balance sheet the 30 year Mortgage with $x_2 = i/i_2 = 62$.

Step 4. The balance of the balance sheet will be $y_0 = x_1 + x_2 + x_3 + x_4 = 373$.

8 Conclusion

We formulate the bank balance sheet allocation problem as a linear programming problem. Using the special features of the problem and linear programming duality we were

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able to derive explicit procedures that have clear financial interpretations. We established a link between the primal balance sheet problem and risk allocation, using duality to compute the prices of credit risk and interest rate risk.

In particular, we found that the optimal balance sheet in the nondegenerate case is a two asset hedge with the possibility to reduce to one asset balance sheet together with the central bank funding in degenerate cases. This property of the optimal balance sheet can easily be combined with a greedy algorithm to handle the practical important case when diversification constraints are involved. We illustrated this method with realistic examples. As usual, the primary solution of the risk allocation problem leads to many directions for further exploration. We elaborate a few below.

In our simplified model we assumed a unlimited supply of riskless central bank funding. In reality this is, of course, not the case. High percentage of short term funding was known to be the primary reason that lead to the fall of Bear Stearns and Lehman Brothers, which triggered the 2008 financial crisis. For this reason, Basel III enforced liquidity restrictions. Adding liquidity risk limitations will increase the type of risks to three. This will make the configuration of optimal balance sheet more interesting and sophisticated. A thorough discussion including the liquidity risk is of great practical importance.

Imposing linearity on risk measures is a reasonable first step. Many important risk measures used in practice such as standard deviation and general coherent risk measures are not linear. The discussion of nonlinear risk measures is a natural next step. Such problems will bear particular practical importance when there are correlations among different types of risk.

Following the pionneering work of Markowitz, portfolio theory is established with a framework of trading off between the expected utility and a risk measure. In the problem discussed here we see the need to dealing with multiple different types of risks. This poses a new challenge to the optimal balance sheet theory on how to deal with not only one risk measure but several. A systematical investigation in this direction seems a worthy journey.

There are of course many other possible directions for further exploration, such as the equilibrium for the distribution of the risk among multiple agents. We hope the research in this paper will stimulate further discussion on the allocation of balance sheet risks by banks, as our framework will allow these financial institutions to better manage their balance sheet risks, and regulators to monitor the incentives of market conditions on the risk profiles of banks.

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